

Data Aggregation Using Evolutionary Computing Based Clustering In Sparse WSN

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ABSTRACT

Wireless Sensor Network composed of self organized wireless sensor nodes distributed in a monitored area collects processes and transmits data acquired from the physical environment. The main goal of the WSN is to aggregate the data and then transmitting that data to multiple sinks in order to reduce energy consumption. Aggregation of data from the nodes is performed based on clusters of the nodes in the network. With the Data Density Correlation Degree (DDCD) clustering method, the sensor nodes are divided into the clusters based on their data density correlation. It is applicable only in the networks where sensor nodes are densely deployed and sampled data change slowly with time. In the paper, we present a method with the usage of mobile agents to implement sparse network design in order to improve packet delivery ratio and accuracy of data aggregation using genetic algorithm for clustering of sensor nodes.

Keywords: Genetic Algorithm; Clustering; Mobile agents; sensor nodes;

INTRODUCTION

Wireless sensor network (WSN) is widely considered as one of the most important technologies for the twenty-first century. In the past decades, it has received tremendous attention from both academia and industry all over the world. A WSN typically consists of a large number of low-cost, low-power, and multifunctional wireless sensor nodes, with sensing, wireless communications and computation capabilities. These sensor nodes communicate over short distance via a wireless medium and collaborate to accomplish a common task, for example,

environment monitoring, military surveillance, and industrial process control. The basic philosophy behind WSNs is that, while the capability of each individual sensor node is limited, the aggregate power of the entire network is sufficient for the required mission.

NETWORK DESIGN CHALLENGES AND ROUTING ISSUES

The design of routing protocols for WSNs is challenging because of several network constraints. WSNs suffer from the limitations of several network resources, for example, energy, bandwidth, central processing unit, and storage.

The design challenges in sensor networks involve the following main aspect:

A. Limited energy capacity

Since sensor nodes are battery powered, they have limited energy capacity. Energy poses a big challenge for network designers in hostile environments, for example, a battlefield, where it is impossible to access the sensors and recharge their batteries. Furthermore, when the energy of a sensor reaches a certain threshold, the sensor will become faulty and will not be able to function properly, which will have a major impact on the network performance. Thus routing protocols designed for sensors should be as energy efficient as possible to extend their lifetime, and hence prolong the network lifetime while guaranteeing good performance overall.

B. Sensor locations

Another challenge that faces the design of routing protocols is to manage the locations of the sensors. Most of the proposed protocols assume that the sensors either are equipped with global positioning system (GPS) receivers or use some localization technique to learn about their locations.

C. Limited hardware resources

In addition to limited energy capacity, sensor nodes have also limited processing and storage capacities, and thus can only perform limited computational functionality. These hardware constraints present many challenges in software development and network protocol design for sensor networks, which must consider not only the energy constraint in sensor nodes, but also the processing and storage capacities of sensor nodes.

D. Massive and random node deployment

Sensor node deployment in WSNs is application dependent and can be either manual or random which finally affects the performance of the routing protocol. In most applications, sensor nodes can be scattered randomly in an intended area or dropped massively over an inaccessible or hostile region. If the resultant distribution of nodes is not uniform, optimal clustering becomes necessary to allow connectivity and enable energy efficient network operation.

NETWORK CHARACTERISTICS AND UNRELIABLE ENVIRONMENT

A sensor network usually operates in a dynamic and unreliable environment. The topology of a network, which is defined by the sensors and the communication links between the sensors, changes frequently due to sensor addition, deletion, node failures, damages, or energy depletion. Also, the sensor nodes are linked by a wireless medium, which is noisy, error prone, and time varying. Therefore, routing paths should consider network topology dynamics due to limited energy and sensor mobility as well as increasing the size of the network to maintain specific application requirements in terms of coverage and connectivity.

A. Data Aggregation

Since sensor nodes may generate significant redundant data, similar packets from multiple nodes can be aggregated so that the number of transmissions is reduced. Data aggregation technique has been used to achieve energy efficiency and data transfer optimization in a number of routing protocols.

B. Diverse sensing application requirements:

Sensor networks have a wide range of diverse applications. No network protocol can meet the requirements of all applications.

EXISTING SYSTEM

The existing system is only useful for applications where sensor nodes are densely deployed. It does not consider the moderated energy consumption when data is transmitted to the sink node. If the network is sparse then packet delivery ratio will be reduced as well as more energy will be consumed. Also, the amount of data conveyed to the sink node can decrease.

PROPOSED SYSTEM

In this we introduce mobile agents who are mainly used to implement the sparse network in order to improve packet delivery ratio as well as accuracy of data aggregation. This is also used to find out the optimized value of the network parameter using evolutionary computing technique for clustering of sensor nodes.

Information processing in WSNs has three major steps namely pre-processing, data aggregation and inference. Pre-processing is the first step of information processing. It includes simple actions performed on raw data such as signal conditioning (cleaning, compression, scaling and etc.), noise filtering and etc. Data aggregation is the process of aggregating data to the fusion center or inference center in WSN. Inference is a process of using machine learning techniques to extract hidden information out of the aggregated data. Most of current researches focus on applying

machine learning algorithms for making inference (step three of information processing in WSNs), such as classifying a moving object in a surveillance WSN based on data gathered by the sensors, abnormal environmental event identification in an environment monitoring.

A. Cluster Evolutionary Algorithm

Input: population size, expecting similarity threshold, mutation probability

Output: Best fit thresholds

1. Randomly generate similarity thresholds between $[0,1]$ with population size
2. Find the Fitness of each Array in population
3. Select the top most half best fit arrays and clone them
4. Do mutation in all of the arrays to get offspring and find fitness again
5. Select top most half best fit arrays from this offspring
6. Merge previous half and current half population to get the next generation
7. Repeat the process till given fitness value $>$ = excepted similarity threshold

One more enhancement is making the clustering as hierarchical clustering to clearly differentiate the generalization from specialization. Child level cluster is crisper in nature than parent level and exhibits more specific nature and

behavior. To do this, same Adaptive Leader follower algorithm is applied to individual cluster to get multi-level. Here adaptive is more significant term because proposed algorithm can handle concept drifts, conceptual similarity, dynamic building of clusters, and dynamic selection of similarity thresholds.

B. Data Mining in Wireless Sensor Networks

One of the major objectives of many WSN research works is to improve or optimize the performance of the entire network in terms of energy-conservation and network lifetime. Most of the research activities focus on the design of efficient routing protocol at the network layer selection of low-power modulation scheme at the physical layer or adoption of power-saving modal of operation at data link layer to achieve energy-awareness in WSNs.

To illustrate how learning is relevant to decentralized inference and to discuss the challenges that WSNs pose, it will be helpful to have a running example at hand. Suppose that the feature space X models the set of measurements observable by sensors in a wireless network. For example, the components of an element $x \in X = \mathbb{R}^3$ may model coordinates in a (planar) environment, and time. $Y = \mathbb{R}$ may represent the space of temperature measurements. A fusion center, or the sensors themselves, may wish to know the temperature at some point in space-time;

to reflect that these coordinates and the corresponding temperature are unknown prior to the network's deployment, let us model them with the random variable (X, Y) . A joint distribution $P_{X,Y}$ may model the spatiotemporal correlation structure of a temperature field. If the field's structure is well understood, i.e., if $P_{X,Y}$ can be assumed known a priori, then an estimate may be designed within the standard parametric framework. However, if such prior information is unavailable, an alternative approach is necessary.

C. Model for Data Mining in WSN using Distributed Learning

Now let us pose a general model for distributed learning that will aid in formulating the problem and categorizing work within the field. Suppose that in a network of m sensors, sensor i has acquired a set of measurements, i.e., training data, $S_i \subset X \times Y$.

Every sensor of the network can read a single value at time and send the data to the Fusion center using network back bone. Later, Distributed learning in WSNs with a fusion center would like to utilize the data which was collected locally to build the overall estimate of the continuously varying field. To achieve this goal divide the network into different clusters and elected the cluster head which is used to collect the data from its members and send the aggregated or summary information to the Fusion Center.

Much of the work in distributed learning differs in the way that the capacity of the links is modeled. The typical assumption is that the topology of these networks is dynamic and perhaps unknown prior to deployment; a fusion center may exist, but the sensors are largely autonomous and may make decisions independently of the fusion center.

RESULTS

This paper had surveyed that Data Mining in WSN application from two perspectives, namely the network associated issue and application associated issue. By using the proposed system the simulation results says that the energy consumption is less when compared to existing system. It also says that the packet delivery ratio is more and it also improves the accuracy of data aggregation, performance of overall Wireless Sensor Network.

CONCLUSION

Adaptive data aggregation with the use of mobile agents and deployment of multiple sinks will reduce energy consumption during the transmission of data in WSN and will also improve the packet delivery ratio as well as accuracy of data aggregation. The clustering procedure can be done with the optimized value of data threshold using evolutionary computing technique such as genetic algorithm. It can work for sparse network design with the introduction of Mobile agents.

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